

Improving ski helmet testing using a biofidelic neck model

ME-482: Biomechanics of the musculoskeletal system
Project 4

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1. Introduction

Protective helmets play an essential role in reducing severe injuries during impacts. In order to ensure an adequate level of safety, several standards have been developed over the years. These standards are generally based on drop tests during which an instrumented dummy head is subjected to different impacts in order to measure the accelerations transmitted to the head. [1]

However, these testing protocols present several limitations. In most cases, the head is considered as a rigid body isolated from the rest of the human body according to the European standard for helmet testing headforms [2]. In real-world impacts, however, the human neck plays a significant role in the overall dynamics of motion. The head does not behave as an isolated rigid body. Instead, it is part of a complex biomechanical system involving the helmet, the head, the neck, and the rest of the body. The response of this system depends on numerous factors, including the deformation of biological tissues, muscle activation, skeletal structures, and nonlinear dynamic interactions between body segments.

The rotational accelerations are particularly important, as they are strongly associated with traumatic brain injuries such as diffuse axonal injury (DAI), which can occur as a result of rapid rotational motion of the brain inside the skull [3].

At the same time, some helmet manufacturers are currently developing their own testing procedures that go beyond the requirements imposed by existing standards. These approaches aim to better reproduce real accident conditions and the injury mechanisms observed in practice [4, 5].

Our project is part of a collaboration with *La Haute école du paysage, d'ingénierie et d'architecture de Genève* (HEPIA). We contacted Professor Breda, who has been working with the drop tower system installed at HEPIA, in order to see if our project could be linked to the improvement of the helmet impact tests set up used at the time by him. The current system used a rigid head and neck system, lacking biofidel components for more accurate testing of the helmets.

In this context, the objective of this project is to develop a simplified mechanical system capable of introducing neck-like compliance into a helmet testing setup. The proposed device is based on a spring architecture designed to allow controlled vertical displacement of the head and to modify the acceleration response during drop tower experiments. Rather than reproducing the complete biomechanics of the human body, the system aims to provide a simplified representation of the neck and its influence during impact. The remainder of the body is assumed to be rigid, as its mass is significantly larger than that of the neck system. Consequently, the proposed device focuses on reproducing localized damping and compliance at the neck level in order to evaluate their effects on the accelerations transmitted to the head during vertical impacts.

Through this approach, the project aims to highlight certain limitations of current testing protocols, which generally consider an isolated rigid head, as well as the potential interest of an approach integrating deformation and energy absorption at the neck level.

2. Current Standards and Limitations

Helmet certification standards are designed to ensure that protective equipment provides an acceptable level of safety during impacts. In the case of ski and snowboard helmets, one of the main reference standards currently used in Europe is the EN 1077 standard. This standard defines several requirements related to shock absorption capacity, penetration resistance, retention systems, and field of vision.

The impact tests specified by EN 1077 are generally performed using an instrumented rigid dummy head equipped with accelerometers. During these tests, a striker with a mass of approximately 5 kg impacts the helmeted headform in order to reproduce impact scenarios represen-

tative of skiing accidents. According to the standard requirements, the measured acceleration transmitted to the head must remain below 250g. See table 1.

Although these testing methods provide a reliable and repeatable framework for certification, they also present several limitations. The headform used during certification is generally considered perfectly rigid and isolated from the rest of the body. As a result, the influence of the neck and upper body dynamics on impact behavior is largely neglected.

This simplification can become problematic when considering real-world injury mechanisms. During an actual accident, the human neck contributes to the transmission, absorption, and redistribution of forces and moments acting on the head.

However, current standards still primarily focus on translational acceleration thresholds and linear impact metrics. While these criteria are effective for reducing skull fractures and severe direct impacts, they are less representative of the complex biomechanical phenomena involved in rotational brain injuries.

For this reason, several helmet manufacturers and research institutions have recently started developing more advanced testing methodologies. Technologies such as MIPS have notably emerged in response to the growing concern surrounding rotationally induced brain injuries. These approaches attempt to better reproduce realistic accident conditions by integrating oblique impacts, rotational measurements, and more biofidelic test setups. Nevertheless, the integration of neck dynamics and energy absorption mechanisms remains limited in most current certification procedures.

Criteria	EN 1077:2007
Category	European ski/snowboard helmet standard (Class A / Class B)
Main objective	Basic protection against impacts and penetration resistance
Impact mass	Instrumented headform ($\sim 3.1\text{--}6.1$ kg depending on size), often approximated as 5 kg
Drop height / energy	1.5 m (shock absorption test) Impact energy $\approx$ 75 J for a 5 kg headform
Approx. impact speed	≈ 5.4 m/s
Impact angle	Linear impacts
Acceleration criterion	Peak acceleration transmitted to headform < 250 g

Table 1: Main characteristics of the EN 1077:2007 ski helmet standard

Source: BS EN 1077:2007 — *Helmets for alpine skiers and snowboarders* [1].

3. Objectives

The objective of this project is to develop and evaluate a simplified mechanical neck system intended to introduce compliant and energy-absorbing behavior into conventional helmet impact testing setups. Unlike current certification methods, which generally consider the head as a rigid isolated body, the proposed system aims to account for part of the effect produced by the human neck during impacts.

More specifically, the project pursues the following objectives:

- Design a mechanical architecture compatible with the existing instrumented dummy head and drop tower constraints;
- Develop a robust and repeatable spring-based system capable of reproducing simplified neck compliance during vertical impacts;
- Quantify the influence of the proposed system on the accelerations transmitted to the dummy head during impact tests;
- Compare the dynamic response obtained with the compliant neck system to that of a conventional rigid mounting configuration;
- Assess whether introducing localized compliance at the neck level could improve the representativeness of helmet testing protocols regarding real-world impact conditions.

The purpose of the proposed device is not to reproduce the complete biomechanics of the cervical spine, but rather to investigate whether a simplified compliant mechanism can alter impact dynamics and potentially reduce peak accelerations experienced by the head. Ultimately, this work aims to explore the relevance of integrating neck-inspired energy absorption into future helmet evaluation methodologies.

4. Mechanical Design

4.1 System Constraints

Several constraints had to be considered during the development of the proposed neck system. First, the instrumented dummy head used for the experiments was already available and could not be modified. The neck mechanism therefore had to remain compatible with the existing head geometry and mounting interfaces.

In addition, the available space between the dummy head and the supporting plate was limited. The neck system consequently had to fit within a maximum height of approximately 20 cm. This geometric constraint strongly influenced the overall architecture and the dimensions of the damping elements.

Another important constraint was related to the geometry of the dummy head. Due to the lower position of the chin compared to the rest of the head, the available vertical displacement of the system is limited. As the neck prototype compresses, the front part of the head may come into contact with the supporting plate relatively early, which had to be considered during the design process.

Manufacturing constraints also played a significant role during the design process. The system was intentionally designed using manufacturing methods readily available at EPFL, mainly laser cutting and additive manufacturing (3D printing). As a result, the geometry of several components had to remain compatible with these fabrication processes in order to simplify manufacturing and assembly.

Finally, the proposed system had to remain sufficiently robust to withstand repeated impacts while preserving a simplified mechanical behavior representative of neck damping during vertical impacts.

4.2 Initial Design Concepts

One of the first concepts considered during the project aimed to reproduce the structure and flexibility of the human neck in a relatively biofidelic manner. The proposed architecture was

based on a stack of aluminum plates intended to represent the vertebrae of the cervical spine. Between each aluminum element, rubber rings were introduced in order to provide flexibility and allow relative motion between the different segments once assembled. The complete structure was then intended to be embedded within a silicone envelope reproducing the surrounding soft tissues.

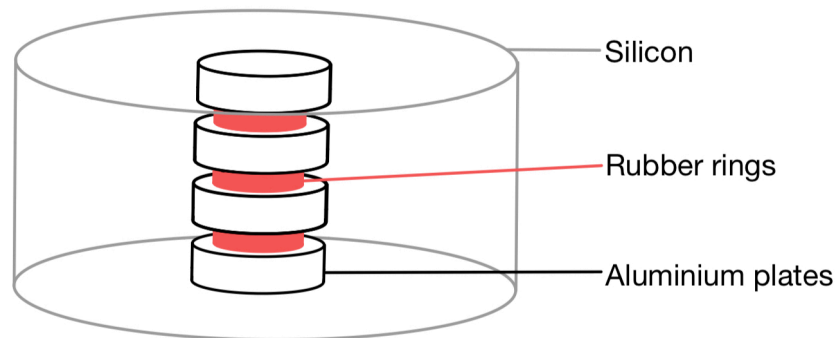


Figure 1: Initial neck concept composed of stacked aluminum parts connected through rubber rings and surrounded by a silicone envelope.

One of the main advantages of this concept was its ability to provide multiple degrees of freedom in three dimensions. This architecture could potentially reproduce bending and rotational motions close to those observed in a real human neck. The combination of rigid and flexible materials also offered a more biomimetic approach, potentially leading to a more realistic global motion of the head during impacts.

However, several important limitations were quickly identified. While the proposed structure could reproduce certain kinematic aspects of neck motion, its mechanical behavior in terms of stiffness and energy absorption remained highly uncertain. Due to the combination of aluminum, rubber, and silicone, the effective force response of the system would have been difficult to predict and characterize accurately.

In addition, concerns were raised regarding the durability and repeatability of the materials used. Discussions with Professor Breda highlighted that the mechanical properties of rubbers and elastomeric materials can evolve significantly after repeated impacts in drop tower experiments. This progressive modification of the material behavior could lead to inconsistent experimental results over time.

The silicone envelope also introduced practical limitations. In the event of tearing or damage during testing, the silicone structure would likely require complete remolding, considerably increasing maintenance time and experimental complexity. Furthermore, the assembly and manufacturing process of such a multi-material structure would have been significantly more difficult compared to a simpler mechanical system.

Another drawback of this approach was the difficulty of controlling and tuning the mechanical parameters of the neck. Properties such as stiffness, damping, and range of motion would strongly depend on nonlinear material behaviors and manufacturing variations. As a result, obtaining repeatable and adjustable experimental conditions would have been challenging.

Despite its limitations, this first concept played an important role in the project by helping identify the key biomechanical features that should be preserved in the final design, particularly the need for controlled flexibility and energy absorption during impacts. [6]

4.3 Final Design Architecture

The final design retained for the project is based on a simplified spring-damper inspired architecture intended to reproduce part of the energy absorption behavior of the human neck during vertical impacts. The system is composed of two main aluminum plates connected by four compression springs distributed symmetrically around the central axis of the structure

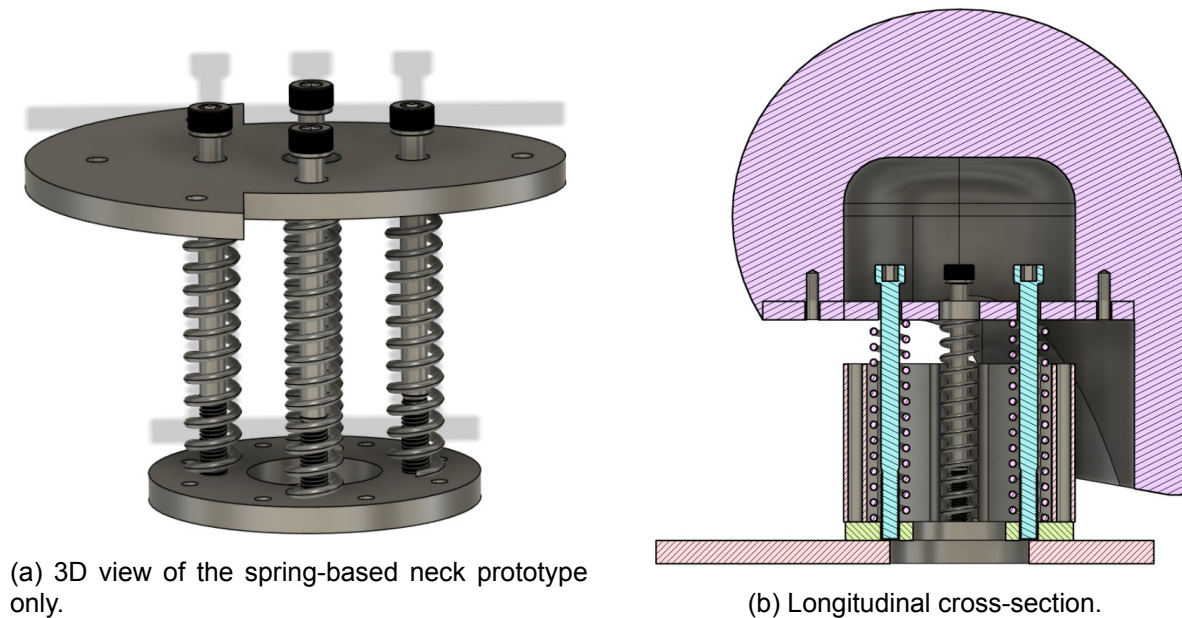


Figure 2: Final neck prototype architecture: external 3D view of the spring-based mechanism and longitudinal cross-section showing the internal guiding and end-stop system.

The upper plate is directly connected to the dummy head assembly, while the lower plate is rigidly attached to the supporting structure of the drop tower. The four springs provide the main compliant behavior of the system and allow a controlled vertical displacement of the head during impacts. Their symmetric arrangement was selected in order to maintain a relatively balanced load distribution and reduce undesired lateral motions.

Guiding elements were also integrated into the design in order to improve the stability of the moving assembly. This guiding structure helps maintain alignment between the upper and lower plates during compression and limits excessive displacement in the horizontal directions. This aspect was particularly important due to the asymmetric geometry of the dummy head, whose front section extends lower than the rear part.

In addition to the spring system, a mechanical end-stop was introduced to limit the maximum admissible displacement of the assembly. This component consists of a cylindrical structure positioned around the springs. The end-stop was designed to prevent excessive compression that could otherwise lead to direct contact between the dummy head and the supporting plate during high-energy impacts.

The cylindrical end-stop structure was manufactured using 3D printing and reinforced using aluminum bars in order to increase its rigidity and impact resistance. This hybrid solution allowed rapid prototyping while maintaining sufficient structural robustness for repeated drop tower experiments.

Compared to the initial biomimetic concepts considered during the early stages of the project, the final architecture intentionally prioritizes simplicity, repeatability, and controllability of the mechanical behavior. Although the proposed system remains a simplified representation of the human neck, it provides a controllable compliant behavior while remaining compatible with the geometric, manufacturing, and experimental constraints of the project.

4.4 Material and Component Selection

- **Aluminum plate:** The upper and lower structural plates were manufactured from aluminum. A plate thickness of 10 mm was required to ensure sufficient rigidity, each structural plate was produced by laser-cutting two identical 5 mm plates and bonding them together.
- **Compression spring:** The four chosen springs (reference LC 120HH 12M) have a free length of 88.9 mm, an outer diameter of 15.88 mm, and a spring rate of 22.3 N/mm. Each spring can withstand a maximum load of approximately 500 N at solid height. The springs are manufactured from music wire and feature squared and ground ends.
- **Mechanical end-stop:** 3D printed (PETG) cylindrical structure reinforced by 8 aluminum bars of 6mm diameter.

5. Biofidelity of the model

5.1 Previous work

Biofidelity is a very wide notion that encompasses, in the context of our project, all the aspects of the neck that need to be modeled for accurate crash testing. Some previous works enlightened the difficulty to establish a biofidel neck model with need for accurate mass, size, width, flexibility among many other characteristics. Moreover, dummies usually used for crash tests are only used once, reflecting a sustainability issue in the dummy market [7].

Several previous works tried to create an accurate dummy neck for experimental use. J.Sparks & al focused their work on the simulation of neck tissue structure [8]. To do so, they focused on silicone as the main material. After quantifying mechanical properties of several silicones, they constructed their model using Ecoflex 0030, which had close shear stress distribution results to porcine muscle after performing characterisation experiments using an indenter. Another previous work we have been inspired from was the one from S.MacGillivray & al. They established a neck model to fit a Hybrid III head for helmet testing. In their model, they used both rigid parts (aluminum), springs, rubber elements and silicone. In that way, they ended up with a quite complete system, biofidelic on many points [6]. But the issue with that set up is first the difficulty of manufacturing and the price of the components. But in the context of this project, the use of silicone was not really accessible. After some research, we realized that it would be difficult to correctly use silicon, which is also an expensive material. Moreover, Professor Breda exposed to us that the idea behind this project would be to have a neck model easy to repair in case of break and long-lasting. Thus, the use of silicone, as well as rubber elements was not the way to go as these materials suffer more from repeated use, especially in crash tests. One element that has been used in some neck models and that is less crumbly than silicone or rubber are springs. The work from T.San Miguel & al proposed a neck made of vertebrae units (supposedly 3D printed) and springs [7]. This model allowed to incorporate the several degrees of freedom neck movement has during crash. As said before, the testing set up of the helmets compel us in some way, and introducing movement of the head in all axes was not easy in that context, as well as not our focus. But springs were still an element that had great potential for introducing compression in the neck model. That's why we focused our model on them.

5.2 Relevance of proposed model

As said above, biofidelity can mean many different things. For the sake of time and feasibility, we decided to actually focus on one biofidel component of the neck : compression. Crash

studies, especially for vehicles, expose the many ways necks can be damaged via crashing : flexion, extension, traction, compression [9]. Since we wanted to focus our model on the use of springs, and because of the testing setup designed with vertical impact on top of the dummy head, compression was one logical element to reproduce.

The main goal of our project was to have a more biofidel neck than the one already used at HEPIA. Thus we had to take into account size of the neck in addition to introducing a correct compression load supported by the neck. The human neck is made of 7 vertebrae, going from C1 to C7. Its length is between 18-19cm [7]. In the present experimental setup, the available space between the dummy head and the supporting plate was limited to approximately 20cm. Although this distance is close to the anatomical length of a human neck, using the full available height was not considered appropriate for the prototype. Since the system is based on compression springs, a rebound of the head-neck assembly may occur after impact. Additional clearance was therefore kept to reduce the risk of unwanted contact between the moving head assembly and surrounding parts during post-impact oscillations. Thus we decided to design a neck representing half of the human neck, from C1 to C5, measuring 10cm. Compression load on the neck is mainly supported by the inter-vertebral discs, D1 to D6, intercalated between the vertebrae from C7 to C2. Previous studies determined the load supported by each disc using cadaver samples, allowing us to evaluate an estimate of the force our system would have to withstand [10] [7]. Vertebral discs between C5 and C1, by summing their individual load supported, can support around 2000N. Thus, the springs used for our model should be able to support a similar load. Moreover, vehicle crash studies established the compression limit for the neck in crashes to be of 4000N [9]. Since we only model half of the neck, it is consistent with 2000N as a limit. This limit suggests that in case of exceeding, the damages to the neck are too important to be accepted (in the contexts of crash testing). That is the reason why we built our model using 4 springs in parallel, each capable of supporting 550N at solid height [11], giving us a total of 2200N.

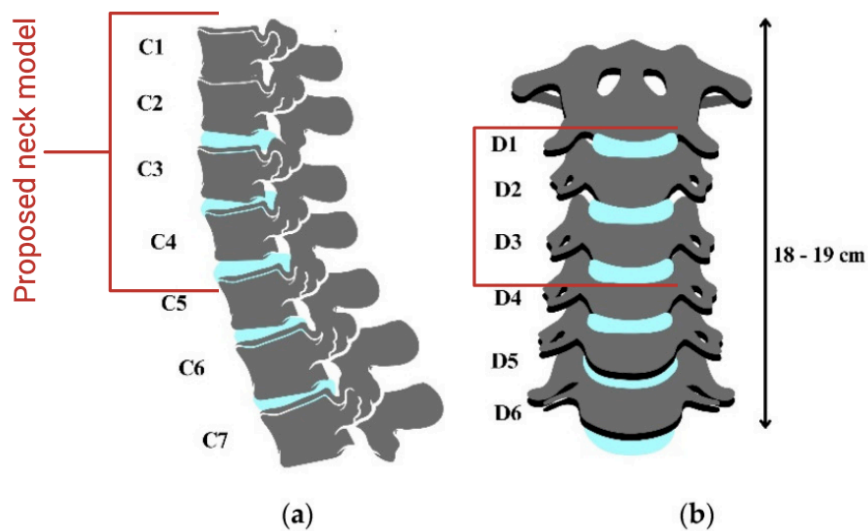


Figure 3: Anatomy of the cervical spine adapted from [7]: (a) lateral view of the 7 cervical vertebrae (C1–C7) with the proposed neck model highlighted in red (C1–C4); (b) frontal view of the intervertebral discs (D1–D6) with the discs included in the proposed model outlined in red.

6. Experimental Protocol

6.1 Drop Tower Setup

The experimental tests were conducted using a vertical drop tower setup. The system consists of an impactor, with a mass of 5 kg in the present experiments, which can be released from different heights. The drop tower has a maximum available height corresponding to approximately five floors, allowing a wide range of potential impact energies.

For the experiments presented in this project, the impactor was positioned at a height of 1 m above the dummy head before being released. The impact then occurred vertically on the helmeted head-neck assembly.

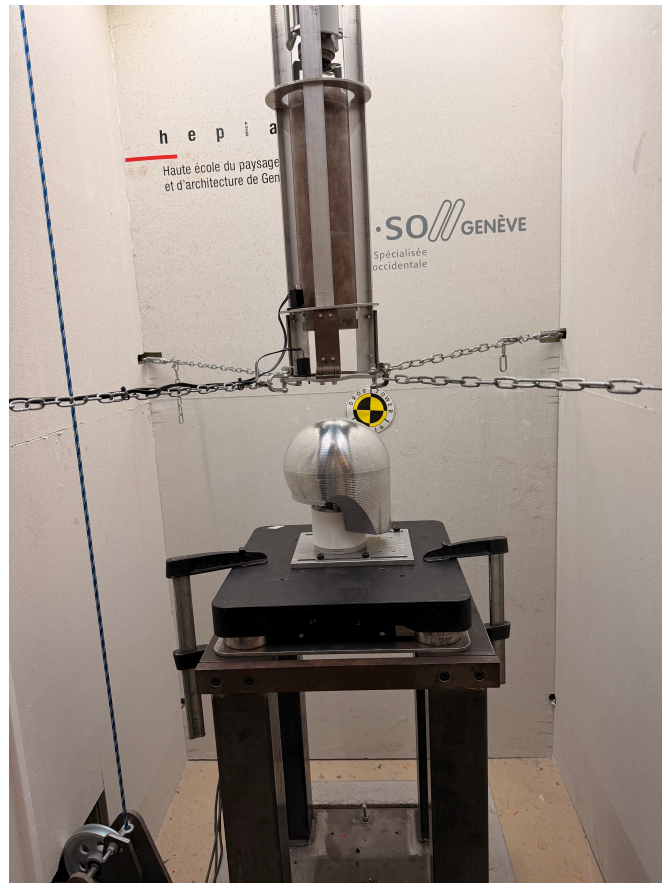


Figure 4: Dummy head and neck prototype mounted in the impact area of the drop tower before vertical impact testing.

6.2 Instrumentation and Data Acquisition

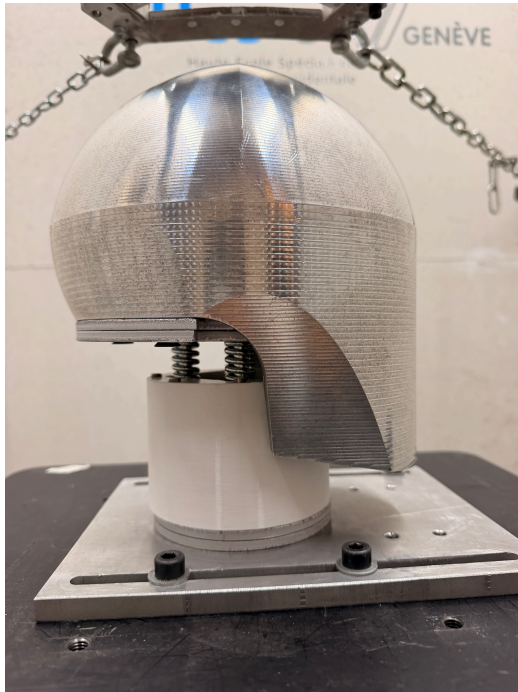
The lower part of the setup includes a force measurement table equipped with four piezoelectric sensors located at its four corners. These sensors allow the resulting impact force transmitted through the test assembly to be measured during each experiment.

The prototype neck system was mounted between the dummy head and the force measurement table. The neck was screwed onto an intermediate plate, which was then rigidly attached to the measurement table in order to ensure a stable mechanical connection during impact.

The impact event was recorded using a high-speed camera. In parallel, the force measurements were acquired at a sampling frequency of 19 200 Hz, allowing the dynamic response of the system to be captured with sufficient temporal resolution.

6.3 Testing Conditions and Procedure

Before the tests, the neck prototype was fixed to the support plate and the assembly was mounted on the force measurement table. The dummy head and helmet were then positioned above the neck system. The impactor was raised to a height of 1 m above the head and released vertically.



(a) Neck prototype mounted under the dummy head.



(b) Drop tower impact setup with helmeted head assembly.

Figure 5: Drop tower setup used for the vertical impact experiments.

During the impact, the force transmitted through the assembly was measured by the piezo-electric sensors, while the high-speed camera recorded the motion of the system. This setup made it possible to observe the mechanical response of the head-neck assembly and to evaluate the influence of the compliant neck prototype on the impact behavior.

The same procedure was intended to be repeated for different configurations in order to compare the response of the system with and without the neck prototype.

6.4 Comparison Methodology

In order to evaluate the influence of the compliant neck prototype on the impact response, a reference test was performed using the rigid neck assembly normally used in the existing experimental setup. This configuration can be considered as a rigid reference case, since no significant relative motion or energy storage is introduced between the dummy head and the supporting structure.

The objective of this comparison was not to provide a fully standardized validation of the prototype, but rather to obtain an order of magnitude of the effect introduced by the spring-based neck system. In particular, the comparison aimed to observe how the force signal evolves during impact and whether the compliant system modifies the peak force, the duration of the impact, or the overall shape of the force-time curve.

Ideally, both tests would have been performed using identical helmets in order to avoid any influence of helmet-to-helmet variability. However, only one helmet was available for the

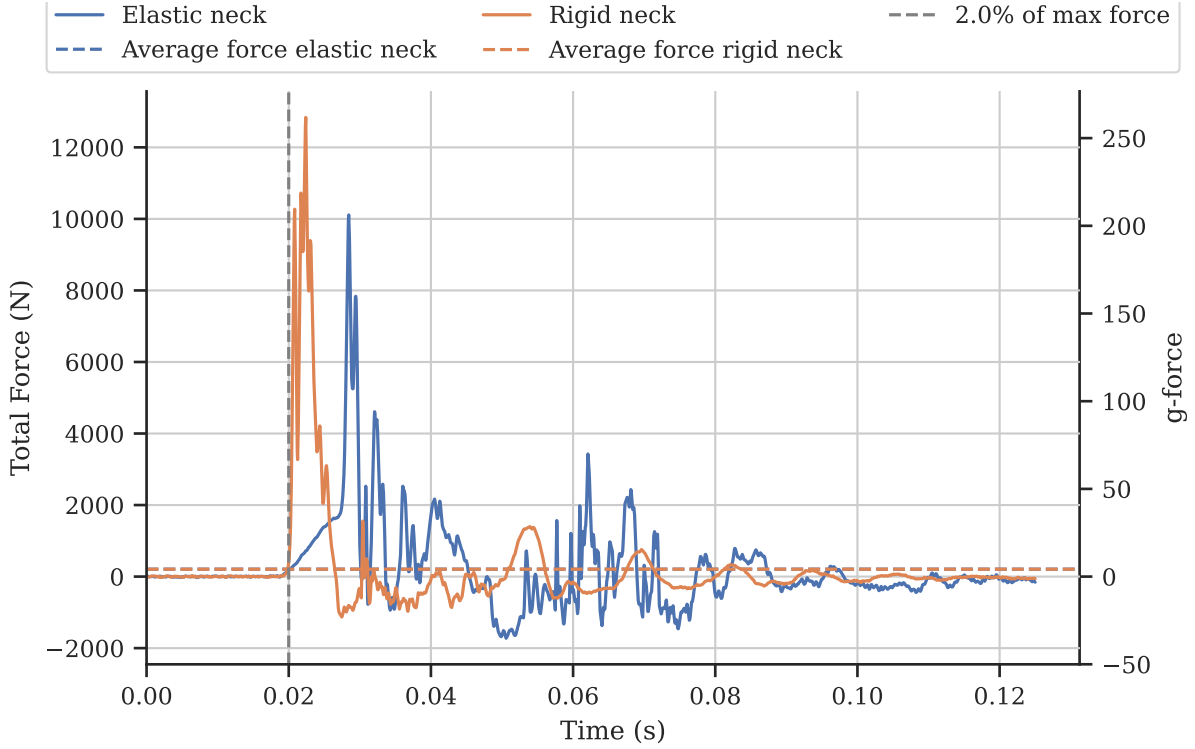


Figure 6: (Force and equivalent G-force measured by the table as a function of time. The rigid neck is shown in orange, while the elastic neck is represented by the blue curve. The signals are synchronized at 2% of the maximum force reached.

experiments. The same helmet was therefore reused for the rigid neck test. This introduces a limitation, since the helmet may have been affected by the first impact and its mechanical response may not be strictly identical during the second test.

Despite this limitation, the comparison remains useful as a experimental assessment of the prototype. By comparing the rigid and compliant configurations under similar test conditions, it becomes possible to identify qualitative trends in the impact response. In particular, the spring-based neck is expected to increase the duration of the impact and reduce the peak force transmitted to the measurement table, while the rigid configuration is expected to produce a shorter and sharper force peak.

7. Results and discussion

The figure shows the time evolution of the total force measured by the force table for the two neck models: an elastic neck and a rigid neck. Both responses exhibit a main impact at approximately $t \simeq 0.02$ s, followed by a transient phase characterized by oscillations.

The rigid-neck model shows a very rapid increase in force, with a maximum peak of approximately 12 to 13 kN. This peak is reached shortly after the beginning of contact, after which the force drops sharply. This behavior indicates an almost instantaneous transmission of the load to the head. The rigid neck absorbs little energy through deformation, which leads to a higher maximum force and a more severe loading during the first instants of the impact.

This severity is also confirmed by the temporal criterion corresponding to the exceedance of $250g$. For the rigid-neck model, the duration during which the acceleration exceeds $250g$ is

$$\Delta t_{250g, \text{rigid}} = 52.08 \mu\text{s}.$$

Although this duration remains very short, it indicates that the rigid model reaches a critical acceleration level for a measurable fraction of the impact.

In contrast, the elastic-neck model exhibits a more gradual response. The force peak is slightly delayed and reaches a lower value, of the order of 10 kN. The elasticity of the neck therefore reduces the maximum force transmitted to the head. In addition, the time spent above the 250g threshold is zero:

$$\Delta t_{250g,elastic} = 0 \mu s.$$

This shows that, in this simulation, the elastic neck prevents the acceleration from exceeding the critical threshold of 250g.

However, this reduction in peak force and maximum acceleration is accompanied by larger and longer-lasting oscillations after the impact, visible up to approximately 0.08 s. This indicates that part of the impact energy is stored and then released by the elastic system.

The comparison of the two models therefore shows that the rigid neck represents a more conservative case in terms of maximum force and maximum acceleration. The elastic neck, on the other hand, attenuates the initial impact by distributing the loading over a longer duration. The main difference between the two responses is therefore not only related to the average force, but mainly to the maximum value, the time at which the peak occurs, and the possible exceedance of the 250g threshold.

Thus, the elastic model appears to better represent realistic biomechanical behavior, since it introduces a certain compliance between the head and the rest of the body. In the case studied, it also appears to be more favorable in terms of impact severity, since it does not exceed the 250g threshold, unlike the rigid model.

8. Possible Improvements

Although the final prototype provides a controlled vertical compliance, several improvements could be introduced in future iterations to make the system more representative of real head-neck dynamics during impacts.

A first improvement would be to include shear and rotational components in the mechanical response of the neck surrogate. The current design mainly allows axial compression along the vertical direction, whereas real head impacts often involve combined translational and rotational motions. These rotational accelerations are particularly relevant because they are associated with shear strains in brain tissue and traumatic brain injuries such as diffuse axonal injury (DAI). One possible way to introduce such effects would be to increase the diameter of the guiding holes around the vertical rods. This would allow a limited lateral displacement of the upper plate during impact.

It would also be important to think about the integration of sensors directly inside the head form. This would make possible to quantify not only the displacement of the neck mechanism but also the dynamic response transmitted to the head. Such measurements would provide more relevant data for evaluating the protective performance of the helmet and for comparing different neck configurations.

9. Conclusion

This project aimed to improve the representativeness of ski helmet impact testing by introducing a simplified biofidelic neck model into a conventional drop tower setup. Current helmet testing standards mainly rely on rigid and isolated headforms, which provide repeatable certification conditions but do not fully account for the influence of the neck on impact dynamics. In this context, the proposed system was designed to introduce controlled compliance and energy absorption between the dummy head and the supporting structure.

A spring-based mechanical architecture was selected as a compromise between biofidelity, simplicity, robustness, and manufacturability. Compared with more complex biomimetic concepts involving silicone or rubber elements, the final design offers improved repeatability, easier maintenance, and tunable mechanical behavior. The use of four compression springs allowed the prototype to reproduce a simplified axial compression response of the cervical region while remaining compatible with the geometric and experimental constraints of the existing HEPIA drop tower setup.

The experimental comparison between the rigid neck configuration and the compliant neck prototype showed that the elastic neck modifies the impact response. In particular, the compliant system reduced the maximum transmitted force and prevented the acceleration from exceeding the 250g threshold in the tested configuration, whereas the rigid neck produced a sharper force peak and briefly exceeded this limit. These results suggest that introducing neck-like compliance can influence helmet impact measurements and may lead to a more realistic assessment of impact severity.

However, the proposed model remains a simplified representation of the human neck. Its response is mainly limited to vertical compression, while real head-neck dynamics involve combined compression, shear, bending, and rotation. In addition, the experimental comparison was limited by the reuse of the same helmet and by the absence of direct acceleration measurements inside the dummy head. Future work should therefore focus on integrating rotational degrees of freedom, improving instrumentation, and performing repeated tests under controlled conditions with multiple helmets.

Overall, this project demonstrates that a simple and robust compliant neck surrogate can be integrated into a helmet testing setup and can significantly affect the measured impact response. Although further development is required before such a system could be used for standardized testing, the results highlight the relevance of considering neck dynamics in future helmet evaluation protocols.

Appendices

A. Group members contribution

The present report has been the result of a group work where we all contributed.

- Kaifun : During the semester, I contributed to the development of the initial design concept by producing sketches and 3D drawings and also participated to the group meetings and discussions. I was then responsible for the CAD modeling of the final neck prototype and the manufacturing and assembly of the final design which included laser cutting, 3D printing, drilling and tapping. I contributed to the writing of the report and I also went twice to Geneva for the project activities related to the drop tower setup and experimental testing.
- Thomas Lepère: During the semester, I contributed to the design of the neck prototype, as well as to its manufacturing, drilling, and assembly. I was also responsible for contacting Prof. Breda and went twice to Geneva: first to visit and understand the drop tower installation, and then to perform the experimental tests. Finally, I took part in the processing and analysis of the experimental data.
- Anne-Laure Thibault : During the semester, I participated in group meetings and discussions related to the project development. At the beginning of the project, I focused on researching the existing literature and contributing to the development of the initial design concepts. Later in the semester, my work focused mainly on investigating and analyzing the standards and regulations applicable to ski helmets. I also visited HEPIA to meet Mr. Breda and observe the drop tower facilities.

- Hortense Liaras : Throughout the semester, I participated to the several discussions we had all together concerning the design of the neck model as well as made some research in the literature. I organized a meeting with Doctor Fournier in order to have his point of view on the project. It is through a discussion with Kaifun that we determined the final design of the neck model. I went 2 times to Geneva, first for discussions with Mister Breda and a visit of the drop tower, second time for performing the testing of our model. The main part I worked on was the biofidelity, searching for litterature references and how we could be more biofidel with our model.

B. Spring Characteristics

Numéro de pièce

LC 120HH 12M

Description	Specification
Spring Type:	Série Compression standard (pouce)
Diamètre extérieur (mm):	15.88
Diamètre du trou (mm):	16.66
Diamètre de la tige (mm):	null
Longueur libre (mm):	88.9
Taux (N/mm):	22.292
Hauteur solide (mm):	63.36
Diamètre du fil (mm):	3.05
Matériau:	Fil de musique
Charge à hauteur solide (N):	552.36
Terminaisons:	Au carré et au sol
Direction du vent:	Option d'usine
Bobines actives:	18.18
Total des bobines:	20.18
Finition:	Plaque de zinc et cuisson au four selon ASTM B633

Figure 7: Spring acquired from Lee Spring [11]

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